Salvatore D. Pace

Curriculum Vitae

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Academic Positions

Institute for Advanced Study — Princeton, NJ
- Member, School of Natural Sciences — Start date: Sept 2026

Education

Massachusetts Institute of Technology — Cambridge, MA
- Ph.D. in Physics — 2026
- Thesis: *When Symmetry Does the Unexpected*
- Advisor: Xiao-Gang Wen

University of Cambridge — Cambridge, UK
- MPhil in Physics — 2021
- Thesis: *Emergent Axions in U(1) Quantum Spin Liquids*
- Advisor: Claudio Castelnovo

Boston University — Boston, MA
- B.A. with honors & M.A. in Physics — 2020
- Thesis: *The Fine Structure Constant in Quantum Spin Ice*
- Advisor: Christopher Laumann

Selected Awards and Honors

MIT Andrew M. Lockett Award for Doctoral Theoretical Physics Research	2026
Kavli Institute for Theoretical Physics Graduate Fellowship	2025
National Science Foundation Graduate Research Fellowship	2022 – 2025
Churchill Scholarship	2020 – 2021
American Physical Society LeRoy Apker Award Finalist	2020
BU College Prize for Excellence in the Physics Department	2020
Learning Assistant of the Year	2019
Goldwater Scholarship	2019

Scientific Papers [Google Scholar]

[20] E. Lew-Smith, S. D. Pace and S.-H. Shao, "Infinite-Order Lattice Chiral Anomalies and CPT," `arXiv:2606.12510`.

[19] S. D. Pace and D. Bulmash, "Lieb-Schultz-Mattis constraints from stratified anomalies of modulated symmetries," `arXiv:2602.11266`.

[18] M. L. Kim, S. D. Pace, and S.-H. Shao, "Symmetry-enforced Fermi surfaces," *Phys. Rev. Lett.* **136** (2026) 176502, [`arXiv:2512.04150`].

[17] S. D. Pace, Ö. M. Aksoy, and H. T. Lam, "Spacetime symmetry-enriched SymTFT: from LSM anomalies to modulated symmetries and beyond," *SciPost Phys.* **20** (2026) 007, [arXiv:2507.02036].

[16] S. D. Pace, M. L. Kim, A. Chatterjee, and S.-H. Shao, 'Parity Anomaly from a Lieb-Schultz-Mattis Theorem: Exact Valley Symmetries on the Lattice," *Phys. Rev. Lett.* **135** (2025) 236501, [arXiv:2505.04684].

[15] S. D. Pace, A. Chatterjee, and S.-H. Shao, "Lattice T-duality from non-invertible symmetries in quantum spin chains," *SciPost Phys.* **18** (2025) 121, [arXiv:2412.18606].

[14] S. D. Pace, H. T. Lam, and Ö. M. Aksoy, "(SPT-)LSM theorems from projective non-invertible symmetries," *SciPost Phys.* **18** (2025) 028, [arXiv:2409.18113].

[13] A. Chatterjee, S. D. Pace, and S.-H. Shao, "Quantized axial charge of staggered fermions and the chiral anomaly," *Phys. Rev. Lett.* **134** (2025) 021601, [arXiv:2409.12220].

[12] S. D. Pace, G. Delfino, H. T. Lam, and Ö. M. Aksoy, "Gauging modulated symmetries: Kramers-Wannier dualities and non-invertible reflections," *SciPost Phys.* **18** (2025) 021, [arXiv:2406.12962].

[11] S. D. Pace and Y. L. Liu, "Topological aspects of brane fields: Solitons and higher-form symmetries," *SciPost Phys.* **16** (2024) 128, [arXiv:2311.09293].

[10] S. D. Pace, C. Zhu, A. Beaudry, and X.-G. Wen, "Generalized symmetries in singularity-free nonlinear σ models and their disordered phases," *Phys. Rev. B* **110** (2024) 195149, [arXiv:2310.08554].

[9] S. D. Pace, "Emergent generalized symmetries in ordered phases and applications to quantum disordering," *SciPost Phys.* **17** (2024) 080, [arXiv:2308.05730].

[8] S. D. Pace and X.-G. Wen, "Exact emergent higher-form symmetries in bosonic lattice models," *Phys. Rev. B* **108** (2023) 195147, [arXiv:2301.05261].

[7] Y.-T. Oh, S. D. Pace, J. H. Han, Y. You, and H.-Y. Lee, "Aspects of $\mathbb{Z}_N$ rank-2 gauge theory in $(2+1)$ dimensions: Construction schemes, holonomies, and sublattice one-form symmetries," *Phys. Rev. B* **107** (2023) 155151, [arXiv:2301.04706].

[6] S. D. Pace and X.-G. Wen, "Emergent higher-symmetry protected topological orders in the confined phase of $U(1)$ gauge theory," *Phys. Rev. B* **107** (2023) 075112, [arXiv:2207.03544].

[5] S. D. Pace and X.-G. Wen, "Position-dependent excitations and UV/IR mixing in the $\mathbb{Z}_N$ rank-2 toric code and its low-energy effective field theory," *Phys. Rev. B* **106** (2022) 045145, [arXiv:2204.07111].

[4] S. D. Pace, C. Castelnovo, and C. R. Laumann, "Dynamical Axions in $U(1)$ Quantum Spin Liquids," *Phys. Rev. Lett.* **130** (2023) 076701, [arXiv:2109.06890].

[3] S. D. Pace, S. C. Morampudi, R. Moessner, and C. R. Laumann, "Emergent Fine Structure Constant of Quantum Spin Ice Is Large," *Phys. Rev. Lett.* **127** (2021) 117205, [arXiv:2009.04499].

[2] S. D. Pace, K. A. Reiss, and D. K. Campbell, "The β Fermi-Pasta-Ulam-Tsingou Recurrence Problem," *Chaos* **29** (2019) 113107, [arXiv:1908.00564].

[1] S. D. Pace and D. K. Campbell, "Behavior and breakdown of higher-order Fermi-Pasta-Ulam-Tsingou recurrences," *Chaos* **29** (2019) 023132, [arXiv:1811.00663].

Invited Talks

Colloquia

University of Minnesota [Slides] Apr '26

Seminars

Bard College [*slides to appear*] Oct '26

University of Minnesota [Slides] May '26

Institute for Advanced Study [Notes] Jan '26

University of Oxford [Notes] Nov '25

Simons Center for Geometry and Physics [Notes], [Recording] Oct '25

CU Boulder CTQM [Slides] Sept '25

Georgia Tech [Slides] May '25

UCLA [Pre-talk notes], [Main talk slides] Feb '25

Symmetry Seminar [Slides], [Recording] Feb '25

Perimeter Institute for Theoretical Physics [Slides], [Recording] Nov '24

Ohio State University [Slides] Oct '24

Harvard [Slides] Oct '24

Boston University [Notes] May '24

Symmetry Seminar [Slides], [Recording] Sept '23

Boston University [Slides] June '22

Max Planck Institute for the Physics of Complex Systems [Slides] Nov '20

Workshops/conferences

Leinweber workshop on Quantum Fields, Matter, and their Symmetries [*Slides to appear*] Dec '26

Symmetries 26 [Slides], [Recording] June '26

GGI Workshop: Defects and Extended Excitations in Quantum Field Theory, Quantum Matter and Statistical Models [Notes], [Recording] June '26

Simons Collaboration Workshop on Confinement [Notes] May '26

OIST TSVP Symposium: Aspects of Generalized Symmetries [Slides] June '25

OIST Thematic Program: Generalized Symmetries in Quantum Matter [Pre-talk notes], [Main talk slides] June '25

KITP Program: Generalized Symmetries in Quantum Field Theory: High Energy Physics, Condensed Matter, and Quantum Gravity [Slides], [Recording] Apr '25

IBS PCS Workshop: Effective Field Theory Beyond Ordinary Symmetries [Slides], [Recording] Dec '24

SCGP Workshop: Applications of Generalized Symmetries and Topological Defects to Quantum Matter [Slides], [Recording] Sept '24

Teaching Experience

Schools and workshops

Invited Lecturer: 2026 Arnold Sommerfeld School on Generalized Symmetries — Fall ’26
[*Notes to appear*]

Invited TA: The Physics and Mathematics of Boundaries, Impurities, and Defects — Fall ’25
[Lecture 1 recording], [Lecture 2 recording], [Notes]

Invited TA: Atlantic TQFT Spring School 2025 — Spring ’25

Massachusetts Institute of Technology

TA: 8.02, Physics II — Spring ’26

Two-time guest lecturer: 8.513, Modern Quantum Many-Body Physics — Fall ’23

Two-time guest lecturer: 8.231, Physics of Solids I — Fall ’22

Boston University

Undergraduate Teaching Assistant

- PY406, Electromagnetic Fields and Waves II — Spring ’20
- PY405, Electromagnetic Fields and Waves I — Fall ’19
- PY452, Quantum Physics II — Fall ’19
- PY451, Quantum Physics I — Spring ’19
- PY410, Statistical Physics & Thermodynamics — Spring ’19
- PY351, Modern Physics I — Fall ’18
- PY313, Waves and Modern Physics — Fall ’18

Guest lecturer: PY410, Statistical Physics & Thermodynamics — Spring ’19

Academic Services

Organizing

Symmetry seminar: Organizer — 2025 – Present

MIT Ultra Quantum Matter seminar: Organizer — 2022 – 2026

MIT physics colloquium committee: Graduate student representative — 2021 – 2024

Mentorship

Mentor for Project SHORT — 2020 – Present

MIT UROP Supervisor — 2022 – 2023

Mentor for Boston University’s PRISM — 2018 – 2020

Journal Referee: SciPost and Physical Review